

Chapter 1

Ideals

1.1 Ideals

Example 1. Consider the set of integer multiples of 3, $3\mathbb{Z}$. $3\mathbb{Z}$ is nonempty since $0 \in 3\mathbb{Z}$; $3a - 3b = 3(a - b) \in 3\mathbb{Z}$ and $3a3b = 9ab = 3(3ab) \in 3\mathbb{Z}$ for any two integers a and b . So $3\mathbb{Z}$ is a subring of $\mathbb{Z}$ and hence a ring (by the subring test, theorem ??).

Now consider any integer r . $r(3a) = (3a)r = 3ar \in 3\mathbb{Z}$. So the subring $3\mathbb{Z}$ “absorbs” any element from its ring $\mathbb{Z}$.

Example 2. The set $T = \{0, 1\}$ is a subring of $\mathbb{Z}$ (by the subring test), but T does not share the “absorbency” property of the example above. As $5 \cdot 1 = 5 \notin T$.

As we move forward, the “absorbency” property which distinguishes these two subrings of $\mathbb{Z}$ will become more important. So we need to give it a name.

Definition 1. A subring A of a ring R is an (two-sided) *ideal* of R if for every $r \in R$ and every $a \in A$ then ra and ar are in A .

Example 1 is an ideal while example 2 is not an ideal.

Of course it will not be necessary to check the property “ ra and ar ” when the ring is commutative (we anyway in example 1). However keep in mind that the subring of a non-commutative ring may still be commutative –in which case the property must be checked.

Example 3. $\mathbb{Z}$ is a subring of $\mathbb{Q}$, but $\mathbb{Z}$ is not an ideal since $3 \cdot \frac{1}{2} = \frac{3}{2} \notin \mathbb{Z}$.

Example 4. $2\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

Proof. This proof is similar to example 1. $2\mathbb{Z}$ is a subring of $\mathbb{Z}$ by the subring test. Let $r \in \mathbb{R}$ and $a \in 2\mathbb{Z}$ so that $a = 2n$ for some $n \in \mathbb{Z}$. $ra = r2n \in 2\mathbb{Z}$. $\square$

- An ideal is a subring that is closed under multiplication with any element from its ring. An ideal is a subring that “absorbs” elements from the rings.

Exercise 1. Prove that 0_R and R are ideals of R for any ring R . The ideal 0_R is called the *zero ideal*.

Exercise 2. Let $\mathbb{R}[x]$, the set of all polynomials with real coefficients. Let C be the subset of $\mathbb{R}[x]$ with constant terms that are 0. For example, $x^3 + x \in C$ but $x^3 + x - 4 \notin C$. Prove that C is or is not an ideal.

As being a subring is a prerequisite of being an ideal, it should not be a surprise that a generalization of the subring test (theorem ??) will simplify the verification of an ideal.

Theorem 1. Ideal test. *A nonempty subset I of a ring R is an ideal if and only if:*

1. *If $a, b \in I$ then $a - b \in I$.*
2. *If $r \in R$ and $a \in I$ then $ra \in I$ and $ar \in I$.*

Proof. ($\Leftarrow$) Let I have properties 1 and 2. I is a ring by the subring test since property 2 is true for all $r \in I \subseteq R$, and property 1 is identical to the property 1 of the subring test. Property 2 also implies the absorbing property of ideals.

($\Rightarrow$) All ideals have property 1 since they are subrings, and property 2 by definition. $\square$

Exercise 3. Prove that the set $I = \{a, 0\} : a \in \mathbb{Z}\}$ is an ideal of the ring $\mathbb{Z} \times \mathbb{Z}$.

Exercise 4. Let A and B be ideals of R . Prove that $A \cap B$ is also an ideal of R .

Exercise 5. Let A and B be ideals of R . Prove that $A \cup B$ is not necessarily an ideal of R .

Exercise 6. If I is an ideal in the ring R and J is an ideal of the ring S , prove that $I \times J$ is an ideal of the ring $R \times S$.

Exercise 7. Let R be a ring with unity 1_R and let I be an ideal of R . If $1_R \in I$ prove $I = R$.

Exercise 8. Let R be a ring with unity 1_R and let I be an ideal of R . If I has an element x and x^{-1} such that $x \cdot x^{-1} = x^{-1} \cdot x = 1_R$ ¹, prove that $I = R$.

Exercise 9. ★ Prove that the set $I_R = \left\{ \begin{bmatrix} x & y \\ 0 & 0 \end{bmatrix} : x, y \in \mathbb{R} \right\}$ is a subring of $M_{2 \times 2}(\mathbb{R})$ that absorbs products on the right but is not an ideal. That is, $ar \in I$ for all $a \in I$ and $r \in M_{2 \times 2}(\mathbb{R})$, but it is possible that $ra \notin I$. Such a subring is called a *right subring*.

Example 1 is an ideal that was formed by multiples of a single element, $3 \in \mathbb{Z}$. Because the entire ideal can be expressed as a product of a finite number of elements, we call such ideals *finitely generated ideals*.

Theorem 2. *Let R be a commutative ring with unity², $c \in R$, and I the set of all multiples of c in R ($I = \{cr : r \in R\}$) Then I is an ideal.*

Exercise 10. ★ Prove theorem 2. Why is the assumption that R has unity necessary?

Definition 2. The ideal in 2 is called the *principal ideal generated by c* , denoted $\langle c \rangle$ ³.

Example 5. Examples 1 and 4 are principal ideals (in $\mathbb{Z}$) generated by 3 and 2 respectively.

Example 6. Recall that the ring of polynomials with integer coefficients is denoted $\mathbb{Z}[x]$. The set generated by multiplying polynomials in $\mathbb{Z}[x]$ with the polynomial x is a principal ideal generated by x (an example of an element in $\langle x \rangle$ is $x(x^2 + 3x) = x^3 + 3x^2$).

Proof. $\mathbb{Z}[x]$ is a commutative ring with unity 1_R . $x \in \mathbb{Z}[x]$ and $\langle x \rangle = \{x \cdot f(x) : f(x) \in \mathbb{Z}[x]\}$ so $\langle x \rangle$ is an ideal by theorem 2. $\square$

Exercise 11. ★ Prove that all ideals in $\mathbb{Z}$ are principal ideals.

- It follows that all ideals in $\mathbb{Z}_n$ are also principal.

Definition 3. A *prime ideal* P of a commutative ring R is a proper ideal⁴ of R such that $a, b \in R$ and $ab \in P$ implies $a \in P$ or $b \in P$.

¹Such an element x is called a *unit*.

²“with unity” is equivalent to “with identity”.

³Also often denoted $\langle c \rangle$

⁴An ideal I in R is proper if $I \neq R$, i.e., I is a proper subset of R .

Example 7. Let $3\mathbb{Z}$ (from example 1) is a prime ideal in $\mathbb{Z}$. For example $27 = 3 \cdot 9 \in 3\mathbb{Z}$ and $3, 9 \notin 3\mathbb{Z}$.

Proof. We know from example 1 that $3\mathbb{Z}$ is a ring, and $3\mathbb{Z} \neq \mathbb{Z}$ ($7 \notin 3\mathbb{Z}$ for example); so $3\mathbb{Z}$ is a proper ideal in $\mathbb{Z}$.

Let $a, b \in \mathbb{Z}$ and suppose that $ab \in 3\mathbb{Z}$. So $ab = 3n$ for some $n \in \mathbb{Z}$ which implies that $3|ab$. Suppose that $b \notin 3\mathbb{Z}$ so that $3 \nmid b$. It follows that $3|a$ since 3 is prime, and so $a \in 3\mathbb{Z}$. $\square$

Exercise 12. Show that $5\mathbb{Z}$ is a prime ideal.

Exercise 13. ★ For an integer greater than 1, $n\mathbb{Z}$ is a prime ideal if and only if n is prime.

Definition 4. A *maximal ideal* M of a commutative ring R is a proper ideal such that whenever I is an ideal of R and $M \subseteq I \subseteq R$ then $I = M$ or $I = R$.

1.2 Congruence and ideals

We are familiar with the idea of congruence. Ideals can be congruent as well.

Example 8. $\mathbb{Z}_4$ has 4 elements and so it has at most 4 ideals since all of its ideals are principal. $\langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 0 \rangle$ are all ideals of $\mathbb{Z}_4$ by theorem 2.

$$\begin{aligned}\langle 1 \rangle &= \{1, 2, 3, 0\} = \mathbb{Z}_4 \\ \langle 2 \rangle &= \{2, 0\} \\ \langle 3 \rangle &= \{3, 2, 1, 0\} \\ \langle 0 \rangle &= \{0\}\end{aligned}$$

So $\langle 1 \rangle = \langle 3 \rangle$. These two ideals are equal. So $\mathbb{Z}_4$ has three distinct ideals, $\langle 1 \rangle, \langle 2 \rangle$, and $\langle 0 \rangle$.

- As these ideals were generated by multiples of elements of $\mathbb{Z}_4$, the distinct ideals will correspond to the divisors of 4.

Exercise 14. Find the distinct ideals of $\mathbb{Z}_{12}$.

Definition 5. Let I be an ideal in a ring R and let $a, b \in R$. Then $a \equiv b \pmod{I}$ if $a - b \in I$. We say “ a is congruent to b modulo I ”.

Example 9. Recall that $a \equiv b \pmod{n}$ means that $n|(a - b)$. Let $I = \langle n \rangle$ be an ideal in $\mathbb{Z}$. $I = \{\dots, -2n, -n, 0, n, 2n, \dots\}$, the set of multiples of n . So we could characterize congruence modulus n as $a \equiv b \pmod{n}$ to mean that $a - b \in I$.

- Congruence modulo n in $\mathbb{Z}$ is equivalent to congruence modulo the ideal $I = \langle n \rangle$.